DATA ANALYTSIS WITH POWER BI

(Airline Service Quality analysis)

Submitted by

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Registration No 12325695

Programme and Section K23MG

Course Code: INT 374

Under the Guidance of

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Discipline of CSE/IT

School of Computer Science

Lovely Professional University, Phagwara

CERTIFICATE

This is to certify that Kothakota Jatin Yadav bearing Registration no. 12325695 has completed INT374 project titled **"Airline Service Quality analysis"** under my guidance and supervision. To the best of my knowledge, the present work is the result of his/her original development, effort and study.

Signature
Name of the Supervisor
Designation of the Supervisor
School of Computer Science
Lovely Professional University
Phagwara, Punjab.

Date: 13/12/2025

DECLARATION

I, Kothakota Jatin Yadav, student of INT 374 under CSE/IT Discipline at, Lovely Professional University, Punjab, hereby declare that all the information furnished in this project report is based on my own intensive work and is genuine.

Date: 13/12/2025

Signature K jatin

Registration No. 12325695

Name of the student Kothakota Jatin Yadav

Airline Service Quality analysis

Acknowledgement

I would like to express my sincere gratitude to my faculty mentor **[Geetika Chatley]**, whose guidance, encouragement, and continuous support played a crucial role in the successful completion of this project. Their valuable suggestions and constructive feedback helped me gain deeper insights into data analytics and business intelligence concepts.

I am also thankful to the **Lovely School of Computer Science**, Lovely Professional University, for providing the necessary resources, academic environment, and learning opportunities that enabled me to work on this project effectively.

I extend my appreciation to my friends and classmates for their cooperation, discussions, and support throughout the project duration. Their inputs and motivation helped me overcome challenges during data preparation, analysis, and dashboard development.

Finally, I would like to thank my family for their constant encouragement, patience, and moral support, which kept me motivated throughout the completion of this project.

1. Introduction

The aviation industry today places strong emphasis on passenger experience and service quality, as these factors directly shape customer loyalty, operational success, and brand perception. With growing air travel demand and diverse passenger expectations, airlines collect large volumes of data related to customer feedback, service performance, delays, travel classes, and demographic profiles. Making sense of this data is essential for airlines to understand passenger needs, address service issues, and enhance overall travel experience.

Business intelligence and data analytics tools help convert raw airline data into clear, meaningful insights. Indicators such as satisfaction scores, service ratings, delay patterns, and customer loyalty levels reveal how passengers perceive different aspects of airline services. When these elements are examined together, they offer a deeper view of how factors like age group, travel class, and service quality influence passenger satisfaction and decision-making.

This project, titled **"Airline Service Quality & Passenger Experience Analysis Dashboard,"** examines airline passenger satisfaction data using **Microsoft Power BI**. The main aim of the project is to build an interactive and user-friendly dashboard that allows users to explore satisfaction trends, evaluate service performance, measure the impact of delays, and identify the key elements shaping passenger experience. Through the use of **DAX calculations, slicers, influencer analysis, and dynamic visuals**, the dashboard enables intuitive and flexible data exploration.

The dashboard is organized into **two analytical sections**. The first section presents a consolidated overview of essential metrics such as overall satisfaction rate, response count, average delays, and service-wise ratings across different customer segments. The second section dives deeper by highlighting influential service factors, comparing performance across travel classes, and uncovering root causes behind passenger satisfaction levels. This structure helps users move smoothly from summary insights to detailed analysis.

Overall, this project showcases how effective data visualization and analytical methods can support the airline industry in making informed, data-driven decisions. It emphasizes the value of interactive dashboards in translating complex passenger data into clear insights that guide service improvements and elevate the customer journey.

2. Dataset Source and Description

The dataset contains airline passenger satisfaction data with attributes related to customer demographics, travel details, service quality ratings, and flight delay information. This dataset captures passenger feedback across multiple service touchpoints, enabling a detailed evaluation of factors influencing overall satisfaction and travel experience.

Key Columns Used:

- Passenger ID
- Gender
- Age
- Age Group (18–29, 30–44, 45–59, 60+)
- Customer Type (Loyal Customer, Disloyal Customer)
- Type of Travel (Business, Personal)
- Travel Class (Business, Eco, Eco Plus)
- Departure Delay (Minutes)
- Arrival Delay (Minutes)
- Satisfaction Flag (Satisfied / Dissatisfied)
- Inflight Service Rating
- Inflight Wi-Fi Service Rating
- Seat Comfort Rating
- Leg Room Service Rating
- Baggage Handling Rating
- Check-in Service Rating
- Online Boarding Rating
- Cleanliness Rating
- Food and Drink Rating
- Gate Location Rating
- On-board Service Rating

3. Dataset Preprocessing

Before developing the Power BI dashboard, the raw airline passenger satisfaction dataset was carefully pre-processed to ensure data accuracy, consistency, and analytical reliability. Dataset preprocessing is a critical step in analytics, as it enhances data quality, supports accurate calculations, and improves the effectiveness of visualizations.

a. Data Cleaning

• Removed duplicate passenger records to prevent repeated counting of responses.

• Handled missing and null values in key columns such as satisfaction status, service ratings, and delay durations.

• Standardized categorical values for fields like customer type, travel class, and type of travel to maintain consistency.

• Corrected inconsistent or invalid entries in service rating columns.

b. Data Type Correction

• Converted age, departure delay, and arrival delay fields into numeric format for accurate calculations.

• Ensured service rating columns were stored as whole numbers suitable for aggregation and comparison.

• Verified satisfaction indicators were correctly formatted for classification and filtering.

c. Data Transformation (Power Query)

• Trimmed extra spaces and cleaned text fields to maintain uniformity.

• Created derived columns such as **Age Group** for demographic-based analysis.

• Categorized satisfaction outcomes into clear labels (Satisfied / Dissatisfied).

• Filtered out invalid or extreme delay values where required to improve analytical reliability.

d. Data Modeling

• Structured the dataset to support smooth interaction between customer attributes, service ratings, and delay metrics.

• Optimized relationships to enable effective cross-filtering across visuals and slicers.

• Designed the data model to support KPI calculations, key influencer analysis, and drill-down insights.

e. Final Dataset Readiness

After preprocessing, the dataset became clean, structured, and analysis-ready. This ensured seamless dashboard development, accurate KPI computation, reliable analytical results, and meaningful insights into passenger satisfaction and airline service performance.

4. DAX (Data Analysis Expressions)

DAX (Data Analysis Expressions) is a powerful formula language used in Power BI to create **measures, calculated columns, and KPIs**. It enables dynamic calculations that automatically update based on filters, slicers, and user interactions within the report.

In this project, DAX was extensively used to calculate **passenger satisfaction metrics, service performance indicators, and delay-related insights**, ensuring accurate and interactive analysis across multiple dimensions.

Key DAX Concepts Used:

• **Measures for dynamic calculations**, preferred over calculated columns for better performance

• **Aggregation functions** such as COUNT, AVERAGE, SUM, and DISTINCTCOUNT to compute KPIs

• **Filter context management** to ensure slicers (age group, travel class, customer type) influence calculations correctly

• **Time and delay-based calculations** for average departure and arrival delays

• **Percentage calculations** to measure satisfaction rate and segment-wise performance

• **Conditional logic (IF, SWITCH)** to categorize satisfaction levels and service ratings

4. Analysis on Dataset

i. General Description

The dataset used in this project consists of thousands of airline passenger feedback records collected across different customer segments and travel conditions. Each record represents an individual passenger's travel experience and captures information related to demographics, travel details, service quality ratings, delays, and overall satisfaction.

The dataset includes the following key attributes:

• Passenger Demographics: Gender, Age, Age Group

• Customer Type: Loyal Customer, Disloyal Customer

• Type of Travel: Business Travel, Personal Travel

• Travel Class: Business, Eco, Eco Plus

• **Service Quality Ratings:**

  o Inflight Service

  o Seat Comfort

  o Leg Room Service

  o Food and Drink

  o Online Boarding

o Check-in Service

o Inflight Wi-Fi

o Cleanliness

o Baggage Handling

o Gate Location

• Delay Metrics: Departure Delay (Minutes), Arrival Delay (Minutes)

• Satisfaction Indicator: Satisfied / Dissatisfied

Each row in the dataset represents a single passenger's response, capturing their service ratings, travel details, delay experience, and satisfaction status. This structured format enables detailed, multi-dimensional analysis using Power BI.

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ii. Specific Requirements

The primary objective of this analysis is to answer key business and operational questions related to airline service quality and passenger experience. The specific requirements addressed through this Power BI dashboard include:

• What factors most strongly influence overall passenger satisfaction?

• How does satisfaction vary across travel class, age group, and customer type?

• Which airline services contribute most to passenger satisfaction or dissatisfaction?

• How do departure and arrival delays impact passenger experience?

• Which customer segments show higher loyalty and satisfaction levels?

These requirements were explored using interactive Power BI visuals such as KPI cards, bar charts, donut charts, key influencer visuals, decomposition trees, and slicers to analyze both high-level trends and deeper insights.

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iii. Analysis Results

Passenger Satisfaction Overview

• Higher Satisfaction Observed Among:

o Business Class passengers

o Loyal customers

o Business travel passengers

- **Lower Satisfaction Observed Among:**
  - Economy class passengers
  - Disloyal customers

This indicates that service quality and loyalty programs play a significant role in shaping passenger satisfaction.

Service Quality Performance

- **Top Performing Services:**
  - Inflight Service
  - Seat Comfort
  - Cleanliness
- **Lower Rated Services:**
  - Inflight Wi-Fi
  - Gate Location

Passengers place higher importance on comfort and onboard experience, while digital and airport-related services show room for improvement.

Impact of Delays on Satisfaction

- Passengers experiencing minimal or no delays reported higher satisfaction levels.
- Increased departure and arrival delays directly correlated with dissatisfaction.

This highlights the critical impact of operational efficiency on customer experience.

Influence of Demographics and Travel Type

- Business travelers reported higher satisfaction compared to personal travelers.
- Middle-aged and frequent traveler's showed higher satisfaction and loyalty.

Customer expectations and travel purpose significantly influence satisfaction perception.

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iv. Data Preprocessing & Visualization in Power BI

Before performing the analysis, the dataset was thoroughly prepared using Power Query and Power Pivot in Power BI to ensure accuracy, consistency, and analytical reliability.

Data Preprocessing Steps:

- Identified and handled missing values in service ratings, delays, and satisfaction fields.
- Removed duplicate and invalid passenger records.
- Standardized categorical fields such as travel class, customer type, and satisfaction status.

- Converted numeric fields (age, delays, ratings) into appropriate data types.
- Verified relationships and measures for accurate aggregation and filtering.

Visualizations Used in the Dashboard

To effectively communicate insights, the following Power BI visuals were implemented:

- **KPI Cards**
  - Total Passengers
  - Satisfaction Rate
  - Average Departure Delay
  - Average Arrival Delay
- **Bar Charts**
  - Service-wise satisfaction ratings
  - Satisfaction by travel class and customer type
- **Donut Charts**
  - Satisfaction distribution (Satisfied vs Dissatisfied)
- **Key Influencer Visual**
  - Factors influencing passenger satisfaction
- **Decomposition Tree**
  - Root cause analysis of satisfaction levels
- **Scatter / Comparative Charts**
  - Delay impact on satisfaction
- **Slicers**
  - Travel Class
  - Age Group
  - Customer Type
  - Type of Travel

These interactive visuals allow users to explore insights dynamically without interacting with raw datasets, making the dashboard intuitive, analytical, and decision-focused.

4. Methodology

Tools Used

- Microsoft Power BI Desktop
- Power Query Editor (for data cleaning and transformation)
- Power Pivot (for data modeling and relationships)
- DAX (Data Analysis Expressions)
- Interactive Visuals, Slicers, and Filters

Data Preparation

• Data Cleaning:

Removed duplicate records, handled missing values, and corrected inconsistent entries using Power Query.

• Data Formatting:

Converted age, delay durations, and service ratings into appropriate numeric formats.

• Data Transformation:

Created derived columns such as age groups and satisfaction categories for better segmentation.

• **Data Modeling:**

Established logical relationships to enable smooth cross-filtering and interactive analysis.

Analysis Approach

• Developed DAX measures to calculate satisfaction rate, average delays, and service performance KPIs dynamically.

• Designed a multi-page interactive Power BI dashboard with KPI cards, bar charts, donut charts, influencer visuals, and decomposition trees.

• Implemented slicers to enable user-driven filtering and exploration.

• Applied cross-visual interactions to support drill-down analysis and comparative insights.

5. Dashboard Design and Features

The Power BI dashboard was designed to be **visually intuitive, interactive, and insight-focused**, enabling users to explore airline passenger satisfaction and service quality data effectively. The layout follows a structured and logical flow, combining key performance indicators with detailed analytical visuals to support data-driven decision-making.

Key Features

• **KPI Cards:**

Used to highlight essential metrics such as total passengers, overall satisfaction rate, average departure delay, average arrival delay, and key service performance scores for quick performance assessment.

• **Bar Charts:**

Display service-wise satisfaction ratings, customer type comparison, and travel class performance to support clear comparative analysis.

• **Donut / Pie Charts:**

Represent the proportion of satisfied versus dissatisfied passengers and class-wise or travel-type distribution, allowing quick understanding of passenger segmentation.

• **Key Influencer Visual:**

Identifies the most impactful factors driving passenger satisfaction and dissatisfaction, such as travel class, service quality, and delay durations.

• **Decomposition Tree:**

Breaks down overall satisfaction into contributing dimensions like age group, customer type, and service ratings for root cause analysis.

• **Scatter Plot:**

Analyzes the relationship between service ratings and satisfaction levels to identify performance patterns and outliers.

• **Slicers:**

Interactive filters for **Age Group, Travel Class, Customer Type, Type of Travel, and Satisfaction Status** enable dynamic exploration of insights.

User Interaction

Users can interact with the dashboard by applying slicers to filter visuals dynamically, compare satisfaction levels across different customer segments, and drill into specific service factors. Cross-filtering between visuals allows deeper insight discovery without interacting with raw data. This interactive design enhances analytical clarity and supports informed, operational and strategic decision-making in the airline industry.

6. Detailed Analysis of Power BI Dashboard

Page 1: Passenger Satisfaction & Service Performance Analysis

1. Overall Passenger Experience (KPIs)

The dashboard presents key performance indicators (KPIs) that provide an immediate overview of airline passenger satisfaction and service quality:

• **Total Passengers Analyzed:** Represents the total number of passenger responses captured in the dataset, reflecting the scale and reliability of the analysis.

• **Overall Satisfaction Rate:**
Shows the percentage of passengers classified as satisfied, offering a direct measure of service performance.

• **Average Departure Delay (Minutes):**
Indicates the average delay experienced by passengers before departure, highlighting operational efficiency.

• **Average Arrival Delay (Minutes):**
Represents delays upon arrival, helping assess punctuality and end-to-end travel experience.

These KPIs are calculated using DAX measures and dynamically update based on slicer selections.

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2. Satisfaction by Travel Class

This visual compares passenger satisfaction across:
• Business Class
• Eco Class
• Eco Plus Class

Key Observations:
• Business Class passengers report the highest satisfaction levels.
• Eco Plus shows moderate satisfaction compared to standard Eco.
• Economy Class has the largest passenger volume but relatively lower satisfaction.

Insight:
Higher service personalization and comfort in premium classes directly influence satisfaction scores.

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3. Satisfaction by Customer Type

This chart highlights satisfaction differences between:
- Loyal Customers
- Disloyal Customers

Key Observations:
- Loyal customers consistently report higher satisfaction levels.
- Disloyal customers show a greater proportion of dissatisfaction.

Insight:
Customer loyalty strongly correlates with positive travel experience and perception of service quality.

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4. Service Quality Ratings Overview

This section evaluates multiple service attributes such as:
- Seat Comfort
- Inflight Wi-Fi
- Food and Drink
- Baggage Handling
- Check-in Service
- Online Boarding
- Cleanliness

Key Observations:
- Seat comfort and onboard service receive higher average ratings.
- Inflight Wi-Fi and gate location score comparatively lower.

Insight:
Improving weaker service touchpoints can significantly enhance overall passenger satisfaction.

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5. Delay Impact on Passenger Satisfaction

This visual analyzes satisfaction levels against departure and arrival delays.

Key Observations:
- Passengers experiencing minimal delays report higher satisfaction.
- Satisfaction declines sharply as delay duration increases.

Insight:
Operational punctuality is a critical driver of passenger experience and perception.

Page 2: Influencing Factors & Deep-Dive Analysis

6. Key Influencers of Passenger Satisfaction

The Key Influencer visual identifies the most impactful factors affecting satisfaction.

Key Observations:
• Travel Class
• Inflight Service Quality
• Seat Comfort
• Customer Type

are the strongest drivers of satisfaction.

Insight:
Service quality factors outweigh demographic factors in shaping passenger experience.

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7. Satisfaction by Age Group

This chart examines satisfaction across different age categories.

Key Observations:
• Middle-aged and older passengers show higher satisfaction levels.
• Younger passengers tend to be more critical of service quality.

Insight:
Passenger expectations vary by age group, requiring targeted service strategies.

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8. Service Rating Comparison Across Travel Classes

This visual compares average service ratings for:
• Business
• Eco Plus
• Eco

Key Observations:
• Business Class consistently outperforms other classes across all service metrics.
• Economy Class shows lower ratings, especially in comfort-related services.

Insight:
Investment in economy-class comfort and services could improve satisfaction at scale.

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9. Overall Service Drivers and Satisfaction Relationship

This combined analysis highlights the relationship between:
• Service Ratings
• Delay Metrics
• Satisfaction Outcome

Key Observations:
• High service ratings can partially offset moderate delays.
• Poor service quality combined with delays results in strong dissatisfaction.

Insight:
Passenger satisfaction is multi-dimensional and influenced by both operational and experiential factors.

Role of Slicers and Interactivity

The dashboard includes interactive slicers for:
• Travel Class
• Customer Type
• Age Group
• Satisfaction Status

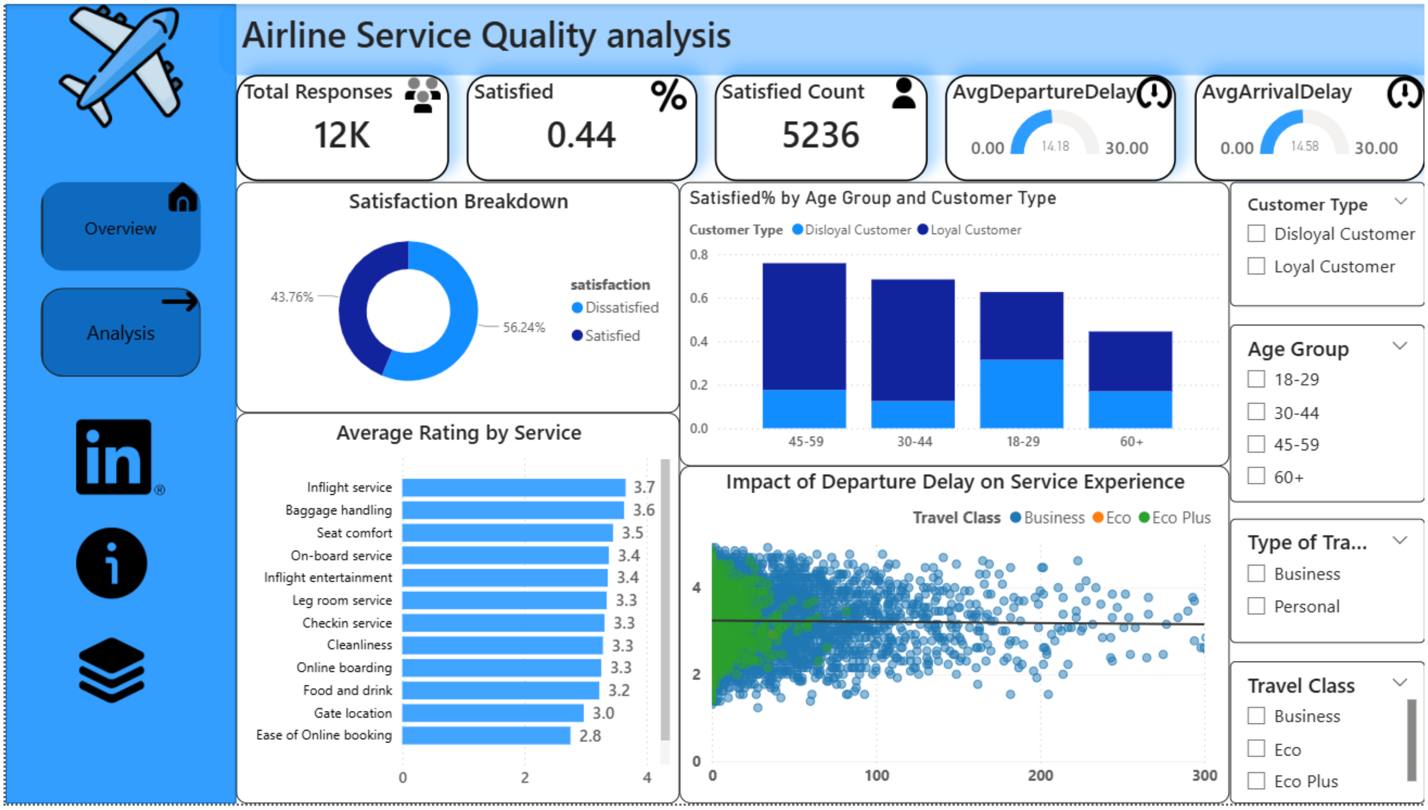
These slicers allow users to dynamically filter visuals, perform comparative analysis, and drill down into specific passenger segments without interacting with raw data, making the dashboard intuitive and decision-focused.

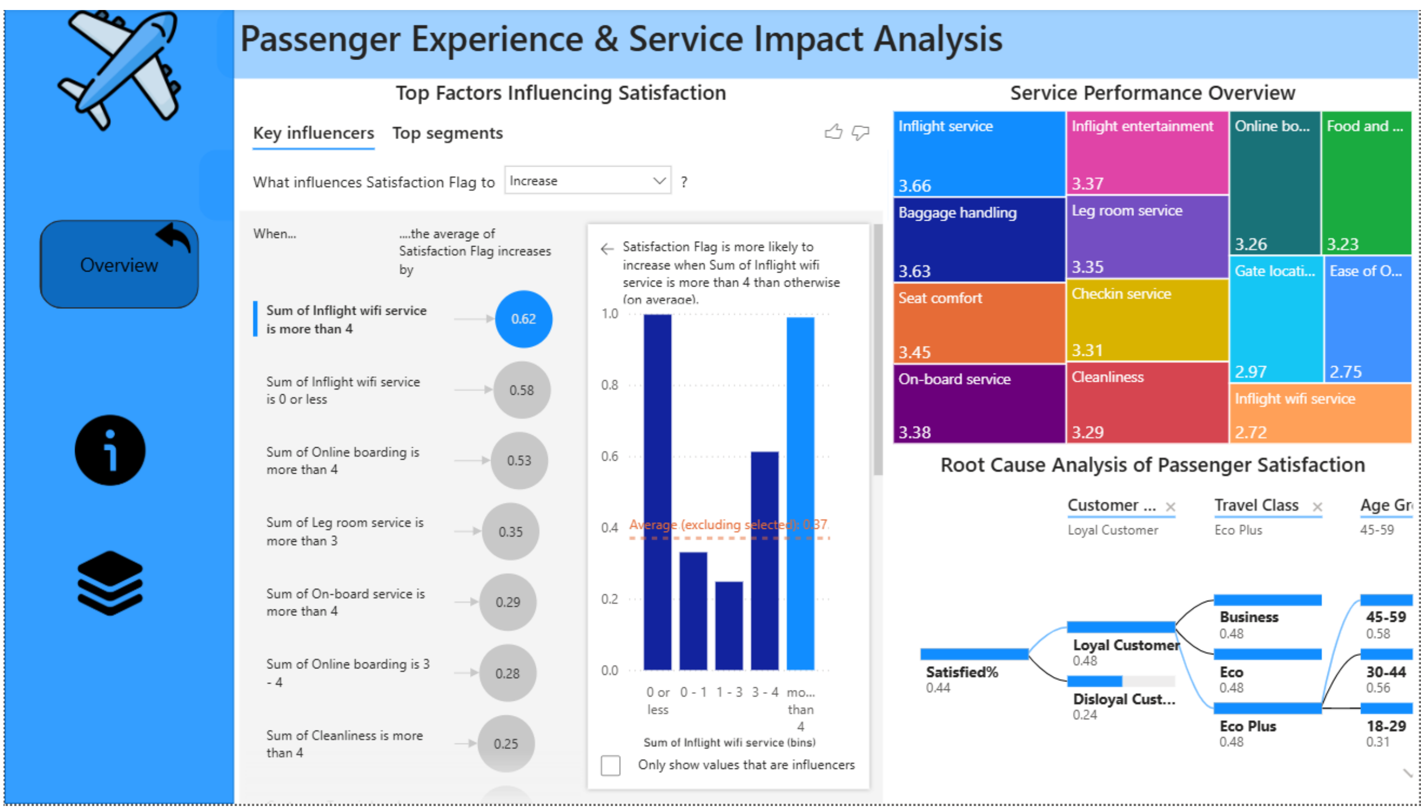
Some Pictures of the dashboard

➔ Example of Data Set

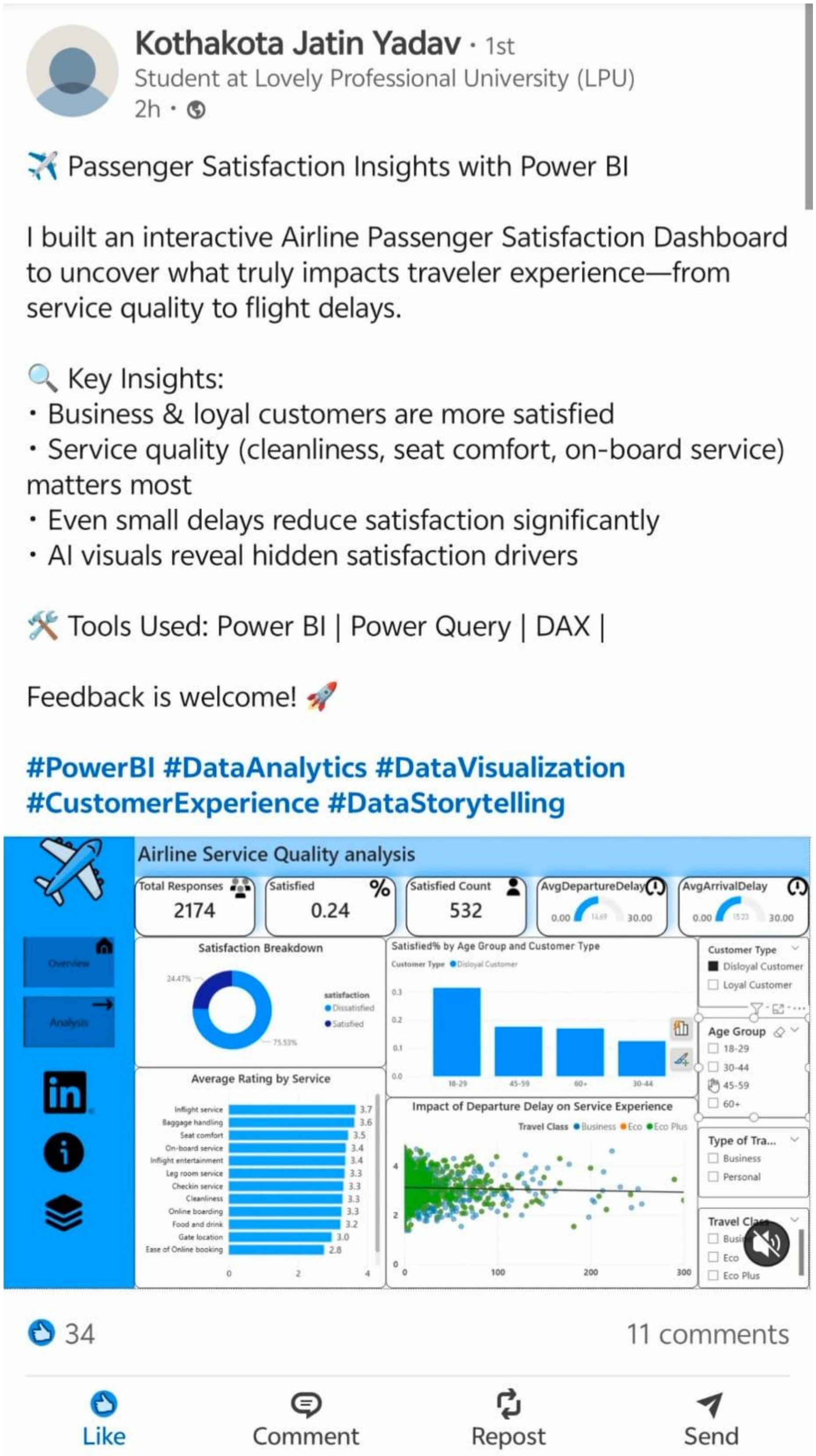
	PassengerFlightKey	Passenger Key	Gender	Age	Travel Key	Class Key
	Valid 100%	Valid 100%	Valid 100%	Valid 100%	Valid 100%	Valid 100%
	Error 0%	Error 0%	Error 0%	Error 0%	Error 0%	Error 0%
	Empty 0%	Empty 0%	Empty 0%	Empty 0%	Empty 0%	Empty 0%
	1000 distinct, 1000 unique	1000 distinct, 1000 unique	2 distinct, 0 unique	71 distinct, 3 unique	2 distinct, 0 unique	3 distinct, 0 unique
1	1	100	Female	52	200	300
2	2	102	Male	20	200	300
3	3	101	Female	36	200	301
4	4	103	Male	44	200	301
5	5	104	Female	49	200	300
6	6	105	Male	16	200	300
7	7	106	Female	77	200	301
8	8	107	Female	43	200	301
9	9	108	Male	47	200	300
10	10	109	Female	46	200	301
11	11	110	Female	47	200	300
12	12	111	Female	33	200	301
13	13	112	Female	46	200	301
14	14	113	Female	60	200	301
15	15	114	Female	52	200	301
16	16	115	Male	50	201	300
17	17	116	Female	31	200	300
18	18	117	Male	52	201	302
19	19	118	Female	43	201	300
20	20	119	Female	50	200	301
21	21	120	Male	60	200	301

	Flight Distance	Inflight wifi service	Departure/Arrival time convenient	Ease of Online booking	Gate location	Food and drink
	Valid 100%	Valid 100%	Valid 100%	Valid 100%	Valid 100%	Valid
	Error 0%	Error 0%	Error 0%	Error 0%	Error 0%	Error
	Empty 0%	Empty 0%	Empty 0%	Empty 0%	Empty 0%	Empty
	678 distinct, 481 unique	6 distinct, 0 unique	6 distinct, 0 unique	6 distinct, 0 unique	5 distinct, 0 unique	6 distinct, 1 unique
1	160	5	4	3	4	
2	192	2	0	2	4	
3	2863	1	1	3	1	
4	3377	0	0	0	2	
5	1182	2	3	4	3	
6	311	3	3	3	3	
7	3987	5	5	5	5	
8	2556	2	2	2	2	
9	556	5	2	2	2	
10	1744	2	2	2	2	
11	1235	4	1	1	1	
12	325	2	5	5	5	
13	1009	5	5	5	5	
14	451	1	1	4	1	
15	925	2	2	2	2	
16	83	3	4	0	3	
17	728	2	5	5	5	
18	1075	5	4	5	3	
19	1927	3	4	3	1	
20	3799	5	5	5	5	
21	612	4	4	4	4	


Airline Service Quality analysis
Total Responses
12K
Satisfied
0.44
Satisfied Count
5236
AvgDepartureDelay
0.00
14.18
30.00
AvgArrivalDelay
0.00
14.58
30.00
Overview
Analysis
Satisfaction Breakdown
43.76%
56.24%
satisfaction
Dissatisfied
Satisfied
Satisfied% by Age Group and Customer Type
Customer Type
Disloyal Customer
Loyal Customer
45-59
30-44
18-29
60+
Customer Type
Disloyal Customer
Loyal Customer
Age Group
18-29
30-44
45-59
60+
Type of Tra...
Business
Personal
Travel Class
Business
Eco
Eco Plus
Average Rating by Service
Inflight service 3.7
Baggage handling 3.6
Seat comfort 3.5
On-board service 3.4
Inflight entertainment 3.4
Leg room service 3.3
Checkin service 3.3
Cleanliness 3.3
Online boarding 3.3
Food and drink 3.2
Gate location 3.0
Ease of Online booking 2.8
Impact of Departure Delay on Service Experience
Travel Class
Business
Eco
Eco Plus


Passenger Experience & Service Impact Analysis
Overview
Top Factors Influencing Satisfaction
Key influencers
Top segments
What influences Satisfaction Flag to
Increase
When...
...the average of Satisfaction Flag increases by
Sum of Inflight wifi service is more than 4 0.62
Sum of Inflight wifi service is 0 or less 0.58
Sum of Online boarding is more than 4 0.53
Sum of Leg room service is more than 3 0.35
Sum of On-board service is more than 4 0.29
Sum of Online boarding is 3 - 4 0.28
Sum of Cleanliness is more than 4 0.25
Satisfaction Flag is more likely to increase when Sum of Inflight wifi service is more than 4 than otherwise (on average).
Average (excluding selected): 0.37
0 or less
0 - 1
1 - 3
3 - 4
mo... than 4
Sum of Inflight wifi service (bins)
Only show values that are influencers
Service Performance Overview
Inflight service 3.66
Inflight entertainment 3.37
Online bo... 3.26
Food and ... 3.23
Baggage handling 3.63
Leg room service 3.35
Seat comfort 3.45
Checkin service 3.31
Gate locati... 2.97
Ease of O... 2.75
On-board service 3.38
Cleanliness 3.29
Inflight wifi service 2.72
Root Cause Analysis of Passenger Satisfaction
Customer ...
Loyal Customer
Travel Class
Eco Plus
Age Gr
45-59
Satisfied% 0.44
Loyal Customer 0.48
Disloyal Cust... 0.24
Business 0.48
Eco 0.48
Eco Plus 0.48
45-59 0.58
30-44 0.56
18-29 0.31

LinkedIn Picture



Kothakota Jatin Yadav · 1st
Student at Lovely Professional University (LPU)
2h ·

✈️ Passenger Satisfaction Insights with Power BI

I built an interactive Airline Passenger Satisfaction Dashboard to uncover what truly impacts traveler experience—from service quality to flight delays.

🔍 Key Insights:
• Business & loyal customers are more satisfied
• Service quality (cleanliness, seat comfort, on-board service) matters most
• Even small delays reduce satisfaction significantly
• AI visuals reveal hidden satisfaction drivers

🛠️ Tools Used: Power BI | Power Query | DAX |

Feedback is welcome! 🚀

#PowerBI #DataAnalytics #DataVisualization #CustomerExperience #DataStorytelling

34
11 comments

Like
Comment
Repost
Send

Conclusion

The Airline Service Quality & Passenger Experience Analysis Dashboard successfully converts complex passenger satisfaction and service performance data into clear, actionable insights using Microsoft Power BI. The analysis highlights that factors such as travel class, customer type, and service quality ratings play a significant role in shaping overall passenger satisfaction. Business-class and loyal customers consistently demonstrate higher satisfaction levels, indicating the strong impact of personalized services and comfort on customer perception.

The dashboard further reveals that service-related factors such as seat comfort, inflight service, cleanliness, and online boarding experience have a greater influence on satisfaction than operational delays alone. While departure and arrival delays affect passenger experience, high-quality onboard and ground services can partially offset dissatisfaction, emphasizing the importance of service excellence across the entire travel journey.

Additionally, the analysis shows that satisfaction patterns vary across age groups, travel purposes, and customer loyalty segments, highlighting the need for targeted service strategies. The use of key influencer analysis enables the identification of the most impactful drivers of satisfaction, allowing airlines to focus improvement efforts on high-impact service areas.

Overall, the interactive dashboard design, effective implementation of DAX-driven KPIs, and dynamic slicers support flexible data exploration and informed decision-making. This project demonstrates strong proficiency in data preprocessing, analytical modeling, and visualization, showcasing how business intelligence tools can be effectively applied to enhance passenger experience and support data-driven strategies within the aviation industry.

6. Future Scope

While this project provides meaningful insights into airline passenger satisfaction and service quality, several enhancements can further strengthen future analyses and expand its real-world applicability:

• Integration with Live Data Sources

Connecting the dashboard to real-time airline operational systems or APIs (flight schedules, delays, customer feedback platforms) would enable continuous monitoring of passenger satisfaction and service performance.

• Predictive Satisfaction Analysis

Applying machine learning models such as classification or regression techniques can help predict passenger satisfaction levels based on service ratings, delays, travel class, and customer demographics.

• Advanced Customer Segmentation

With additional customer-level data, deeper segmentation based on travel frequency, loyalty status, and behavioral patterns can be performed to personalize services and improve retention strategies.

• Operational Performance KPIs

Incorporating advanced KPIs such as on-time performance rate, service recovery effectiveness, complaint resolution time, and customer lifetime value can provide a more holistic performance evaluation.

• Sentiment Analysis on Feedback

If textual passenger reviews or comments are available, natural language processing (NLP) techniques can be used to analyze sentiment and uncover hidden customer perceptions.

• Automation and Scalability

Automating data refresh, preprocessing, and report generation using Power BI Service, scheduled refreshes, or Python-based pipelines can improve efficiency and support large-scale deployment.

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